# Lab 2 - Configuring Basic Single-Area OSPFv2

**Answer Sheet**

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## Part 2: Configure and Verify OSPF Routing

### Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table? 2

show ip route ospf

## Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? Yes 2

Does R2 show up as an OSPF neighbor on R3? No 2

What does this information tell you?

All traffic to the 192.168.2.0/24 network from R3 will be routed through R1. The S0/0/1 interface on R2 is still configured as a passive interface, so OSPF routing information is not being advertised across this interface. The 129 accumulated cost results from the fact that traffic from R3 to the 192.168.2.0/24 network must pass through two T1 (1.544 Mb/s) serial links (at a cost of 64 each) plus the R2 Gigabit 0/0 LAN link (at a cost of 1). 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

2 65 (One T1 (1.544 Mb/s) serial link (at a cost of 64) plus the R2 Gigabit 0/0 LAN link (at a cost of 1)).

Is R2 listed as an OSPF neighbor to R3? Yes 2

## Part 5: Change OSPF Metrics

### Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Today's equipment supports link speeds that are faster than 100 Mb/s. To obtain a more accurate cost calculation for these faster links, a higher default reference-bandwidth setting is needed.

### Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24: R1 S0/0/1 + R3 G0/0 (781+1=782). The cost to 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 (781+64=845).

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562. Each serial link now has a cost of 781, and the route to the 192.168.23.0/24 network travels over two serial links. 781 + 781 = 1,562.

### Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, or 781 + 781 + 1 = 1,563. This metric is smaller than the accumulated cost of R1-S0/0/1 + R3-G0/0, or 1565 + 1 = 1,566.

## Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Router ID assignments control the Designated Router (DR) and Backup Designated Router (BDR) election process on a multi-access network. If the router ID is associated with an active interface, it can change if the interface goes down. For this reason it should be set using the IP address of a loopback interface (which cannot go down) or set using the router-id command. 4

2. Why is the DR/BDR election process not a concern in this lab?

   The DR/BDR election process is only an issue on a multi-access network such as Ethernet or Frame Relay. The serial links used in this lab are point-to-point links, so no DR/BDR election is performed. 4

3. Why would you want to set an OSPF interface to passive?

   4

   Configuring a LAN interface as passive eliminates unnecessary OSPF routing information on that interface, freeing up bandwidth. The router will still advertise the network to its neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF's use of areas (with a backbone Area 0 and non-backbone areas) makes it hierarchical and scalable by segmenting the link-state database and limiting LSA flooding. 4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   (router ospf <process-id>)

   network 10.0.0.0 0.255.255.255 area 0 4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The router will use the EIGRP route because it has the lowest administrative distance (90), compared to OSPF (110) and RIP (120). When routes come from different protocols, administrative distance, not metric, determines the preferred route. 4